IDX G9 MATH H+ STUDY GUIDE ISSUE S2 FINALS

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This guide is a combined guide. Due to the final range including content from before the monthlies, we have included content from the previous issue in this guide for your convenience.

Chapter 3 Exponential and Logarithmic Functions

3.1 Powers and Exponents

Properties (given $a, b \in \mathbb{R}, m, n \in \mathbb{Z}^+$)

-multiplication: $a^m \times a^n = a^{m+n}$

-division with the same base: $\frac{a^m}{a^m} = a^{m-n}$

-division with the same power: $\frac{a^m}{b^m} = \left(\frac{a}{b}\right)^m$

-power of a power with the same base: $(a^m)^n = a^{mn}$

-power of a power with the same power: $a^m \times b^m = (ab)^m$

-negative one to a power: $(-1)^n = \begin{cases} -1 & n = 2k-1 \\ 1 & n = 2k \end{cases}, k \in \mathbb{Z}$

-zero power: $a^0 = 1\ (a \neq 0)$

-negative index: $a^{-m} = \frac{1}{a^m}\ (a \neq 0)$

-rational index: $a^{\frac{m}{n}} = \sqrt[n]{a^m}$

If n is even, then $a^m \geq 0$. If n is odd, then $a^m \in \mathbb{R}$.

Extended range of index to real numbers

For $a, b > 0$ and $s, t \in \mathbb{R}$, we still have

$$a^s \times a^t = a^{s+t}, (a^s)^t = a^{st}, (ab)^t = a^t b^t$$

3.2 Logarithms

$$y = b^N \Leftrightarrow N = \log_b y$$

N is the logarithm of *y* to base *b*.

Definition

$$y = b^N \Leftrightarrow N = \log_b y\,, y > 0, b > 0, b \neq 1$$

Special notations:

$$\log_{10} x = \log x\ (\lg x)$$

$$\log_e x = \ln x\,, e = \lim_{x \to \infty} \left(1 + \frac{1}{x}\right)^x$$

Law of Logarithms

For $x, y > 0, a > 0, a \neq 1$, we have

-addition: $\log_a x + \log_a y = \log_a xy$

-difference: $\log_a x - \log_a y = \log_a \left(\frac{x}{y}\right)$

Taking logarithms from both sides

$$x = y \Leftrightarrow \log_a x = \log_a y$$

Change of Base

$$\log_a b = \frac{\log_k b}{\log_k a}, a, k \in \mathbb{R}^+/\{1\}$$

$$\log_a b = \frac{1}{\log_b a}$$

Other properties

For $a > 0, a \neq 1, x > 0,$

-$\log_a a = 1$

-$\log_a 1 = 0$

-$\log_a x^{-1} = -\log_a x$

-$\log_{\frac{1}{a}} x = -\log_a x$

-$a^{\log_a x} = x$

-$\log_{a^M} b^N = \frac{N}{M} \log_a b$

3.3 Exponential and logarithmic functions

Supplementary notice:

Any function $y = \frac{ax+b}{cx+d}$ $(ad \neq bc)$ can be translated into the function $y = \frac{k}{x}$ $(k \neq 0)$.

Exponential functions

$$f(x) = b^x \ (b > 0, b \neq 1)$$

Properties:

$f(x) = b^x$	$b \in (0,1)$	$b \in (1, \infty)$
Domain	$\mathbb{R}$	$\mathbb{R}$
Range	$(0, \infty)$	$(0, \infty)$
Even/Odd	Neither	Neither
Monotonicity	Strictly decreasing	Strictly increasing

Asymptote	$y = 0$	$y = 0$

Logarithmic functions

$$f(x) = \log_b x \ \ (b > 0, b \neq 1)$$

$f(x) = \log_b x$	$b \in (0,1)$	$b \in (1, \infty)$
Domain	$(0, \infty)$	$(0, \infty)$
Range	$\mathbb{R}$	$\mathbb{R}$
Even/Odd	Neither	Neither
Monotonicity	Strictly decreasing	Strictly increasing
Asymptote	$x = 0$	$x = 0$

3.4 Exponential and logarithmic equations

Types of exponential equations:

1. $a^x = b \ (a > 0, a \neq 1, b > 0) \Leftrightarrow x = \log_a b$
2. $a^{f(x)} = a^{\phi(x)} (a > 0, a \neq 1) \Leftrightarrow f(x) = \phi(x)$
3. $A \cdot a^{2x} + B \cdot a^x + C = 0 \ (a > 0, a \neq 1)$
4. $A \cdot a^{2x} + B \cdot a^x \cdot b^x + C \cdot b^{2x} = 0 \ (a > 0, a, b \neq 1, a \neq b)$

Types of logarithmic equations:

1. $A \cdot \log_a^2 x + B \cdot \log_a x + C = 0 \ (a > 0, a \neq 1)$
2. $\log_a f(x) = \log_a \phi(x) \ \ (a > 0, a \neq 1) \Leftrightarrow \begin{cases} f(x) > 0 \\ \phi(x) > 0 \\ f(x) = \phi(x) \end{cases}$

Chapter 5 Trigonometry

5.1 Measure of angles

Radian Measure: radian measure of a central angle is the number of radius units in the length of the corresponding arc.

Conversion formulas: $\frac{\alpha}{180} = \frac{\theta}{\pi}, l = \theta r, S = \frac{1}{2} lf = \frac{1}{2}\theta r^2$

An angle is formed by rotating a ray about the vertex to another ray.

If the rotation is counterclockwise, the angle is positive.

If the rotation is clockwise, the angle is negative.

Coterminal angles

Angles of different measure in standard position but having the same terminal side.

$$\{x | x = \theta + 2k\pi, k \in \mathbb{Z}\}$$

Quadrantal angles

$$\theta = \frac{k\pi}{2}, k \in \mathbb{Z}$$

5.2 Trigonometric ratio of arbitrary angles

Let θ be an angle in standard position, and let (x, y) be a point distinct form the origin on the terminal ray of θ. Then

$$\cos\theta = \frac{x}{r}, \sin\theta = \frac{y}{r}, \tan\theta = \frac{y}{x}, \text{ where } r = \sqrt{x^2 + y^2}$$

Unit circle

The terminal ray of an angle θ in standard position intersects the unit circle at a unique point $P(x, y)$ where $\mathrm{x} = \cos\theta$ and $y = \sin\theta$.

Basic properties:

1. $\sin\theta \in [-1,1], \cos\theta \in [-1,1], \tan\theta \in \mathbb{R}$
2. $\begin{cases} \sin\theta > 0 \text{ when } \theta \in \mathrm{I}, \mathrm{II} \\ \sin\theta < 0 \text{ when } \theta \in \mathrm{III}, \mathrm{IV} \end{cases}$

$$\begin{cases} \cos\theta > 0 \text{ when } \theta \in \mathrm{I, IV} \\ \cos\theta < 0 \text{ when } \theta \in \mathrm{II, III} \end{cases}$$

$$\begin{cases} \tan\theta > 0 \text{ when } \theta \in \mathrm{I, III} \\ \tan\theta < 0 \text{ when } \theta \in \mathrm{II, IV} \end{cases}$$

3. $\sin(\theta + 2k\pi) = \sin\theta, \cos(\theta + 2k\pi) = \cos\theta, \tan(\theta + 2k\pi) = \tan\theta$

Relations

1. $\tan\alpha \cdot \cot\alpha = 1, \sin\alpha \cdot \csc\alpha = 1, \cos\alpha \cdot \sec\alpha = 1$
2. $s\frac{\sin\alpha}{\cos\alpha} = \tan\alpha = \frac{\sec\alpha}{\csc\alpha}, \frac{\cos\alpha}{\sin\alpha} = \cot\alpha = \frac{\csc\alpha}{\sec\alpha}$
3. $\sin^2\alpha + \cos^2\alpha = 1, 1 + \tan^2\alpha = \sec^2\alpha, 1 + \cot^2\alpha = \csc^2\alpha$

Induction Formulas (奇变偶不变，符号看象限)

1. $\sin(\alpha + 2k\pi) = \sin\alpha, \cos(\alpha + 2k\pi) = \cos\alpha, \tan(\alpha + 2k\pi) = \tan\alpha$
2. $\sin(-\alpha) = -\sin\alpha, \cos(-\alpha) = \cos\alpha, \tan(-\alpha) = -\tan\alpha$
3. $\sin(\pi - \alpha) = \sin\alpha, \cos(\pi - \alpha) = -\cos\alpha, \tan(\pi - \alpha) = -\tan\alpha$
4. $\sin(\pi + \alpha) = -\sin\alpha, \cos(\pi + \alpha) = -\cos\alpha, \tan(\pi + \alpha) = \tan\alpha$
5. $\sin\left(\frac{\pi}{2} - \alpha\right) = \cos\alpha, \cos\left(\frac{\pi}{2} - \alpha\right) = \sin\alpha, \tan\left(\frac{\pi}{2} - \alpha\right) = \cot\alpha$
6. $\sin\left(\frac{\pi}{2} + \alpha\right) = \cos\alpha, \cos\left(\frac{\pi}{2} + \alpha\right) = -\sin\alpha, \tan\left(\frac{\pi}{2} + \alpha\right) = -\cot\alpha$
7. $\sin\left(\frac{3\pi}{2} - \alpha\right) = -\cos\alpha, \cos\left(\frac{3\pi}{2} - \alpha\right) = -\sin\alpha, \tan\left(\frac{3\pi}{2} - \alpha\right) = \cot\alpha$
8. $\sin\left(\frac{3\pi}{2} + \alpha\right) = -\cos\alpha, \cos\left(\frac{3\pi}{s} + \alpha\right) = \sin\alpha, \tan\left(\frac{3\pi}{2} + \alpha\right) = -\cot\alpha$

5.3 Formulas in trigonometry

For equation $\cos\theta = a$ or $\sin\theta = b$, the terminal ray of θ can be determined by $P(a, b)$, the intersection of the terminal ray of θ with the unit circle.

Addition and difference formula

$$\cos(\alpha \pm \beta) = \cos\alpha\cos\beta \mp \sin\alpha\sin\beta$$

$$\sin(\alpha \pm \beta) = \sin\alpha\cos\beta \pm \cos\alpha\sin\beta$$

$$\tan(\alpha \pm \beta) = \frac{\tan\alpha \pm \tan\beta}{1 \mp \tan\alpha\tan\beta}$$

Reference angle formula

For $a, b, x \in \mathbb{R}$,

$$a\sin x + b\cos x = \sqrt{a^2+b^2}\left(\frac{a}{\sqrt{a^2+b^2}}\sin x + \frac{b}{\sqrt{a^2+b^2}}\cos x\right) = \sqrt{a^2+b^2}\sin(x+\phi).$$

Where $\sin\phi = \frac{b}{\sqrt{a^2+b^2}}$ and $\cos\phi = \frac{a}{\sqrt{a^2+b^2}}$

It can also be expressed as $a\sin x + b\cos x = \sqrt{a^2+b^2}sin\left(x + \arctan\left(\frac{b}{a}\right)\right)$

Double angle formula

$$\sin 2\alpha = 2\sin\alpha\cos\alpha$$

$$\cos 2\alpha = \cos^2\alpha - \sin^2\alpha = 1 - 2\sin^2\alpha = 2\cos^2\alpha - 1$$

$$\tan 2\alpha = \frac{2\tan\alpha}{1-\tan^2\alpha}$$

Triple angle formula

$$\sin 3\alpha = 3\sin\alpha - 4\sin^3\alpha = 4\sin\alpha\sin\left(\frac{\pi}{3}+\alpha\right)\sin\left(\frac{\pi}{3}-\alpha\right)$$

$$\cos 3\alpha = 4\cos^3\alpha - 3\cos\alpha = 4\cos\alpha\cos\left(\frac{\pi}{3}+\alpha\right)\cos\left(\frac{\pi}{3}-\alpha\right)$$

$$\tan 3\alpha = \frac{3\tan\alpha - \tan^3\alpha}{1-3\tan^2\alpha} = \tan\alpha\tan\left(\frac{\pi}{3}+\alpha\right)\tan\left(\frac{\pi}{3}-\alpha\right)$$

Omnipotent formula

$$\tan\alpha = \frac{2\tan\frac{\alpha}{2}}{1-\tan^2\frac{\alpha}{2}}$$

$$\sin\alpha = \frac{2\tan\frac{\alpha}{2}}{1+\tan^2\frac{\alpha}{2}}$$

$$\cos\alpha = \frac{1-\tan^2\frac{\alpha}{2}}{1+\tan^2\frac{\alpha}{2}}$$

Half angle formula

$$\sin\frac{\alpha}{2} = \pm\sqrt{\frac{1-\cos\alpha}{2}}$$

$$\cos\frac{\alpha}{2} = \pm\sqrt{\frac{1+\cos\alpha}{2}}$$

$$\tan\frac{\alpha}{2} = \pm\sqrt{\frac{1-\cos\alpha}{1+\cos\alpha}} = \frac{1-\cos\alpha}{\sin\alpha} = \frac{\sin\alpha}{1+\cos\alpha}$$

Sum-to-product formula

$$\sin\alpha + \sin\beta = 2\sin\left(\frac{\alpha+\beta}{2}\right)\cos\left(\frac{\alpha-\beta}{2}\right)$$

$$\cos\alpha + \cos\beta = 2\cos(\frac{\alpha+\beta}{2})\cos(\frac{\alpha-\beta}{2})$$

$$\sin\alpha - \sin\beta = 2\cos(\frac{\alpha+\beta}{2})\sin(\frac{\alpha-\beta}{2})$$

$$\cos\alpha - \cos\beta = -2\sin(\frac{\alpha+\beta}{2})\sin(\frac{\alpha-\beta}{2})$$

Product-to-sum formula

$$\sin\alpha\cos\beta = \frac{1}{2}[\sin(\alpha+\beta) + \sin(\alpha-\beta)]$$

$$\cos\alpha\sin\beta = \frac{1}{2}[\sin(\alpha+\beta) - \sin(\alpha-\beta)]$$

$$\cos\alpha\cos\beta = \frac{1}{2}[\cos(\alpha+\beta) + \cos(\alpha-\beta)]$$

$$\sin\alpha\sin\beta = -\frac{1}{2}[\cos(\alpha+\beta) - \cos(\alpha-\beta)]$$

Unit 5 Trigonometry

5.4 Sine and cosine functions

Periodic function: there is a positive number p, which is the period of f, such that $f(x+p)=f(x)$ for all x in the domain of f.

Fundamental period: smallest period of a periodic function

Amplitude (A): $\frac{1}{2}(M-m)$

Sine curve:	$y = \sin x$
Domain	$\mathbb{R}$
Range	$[-1,1]$
Parity	Odd
Periodicity	Periodic function with fundamental period of 2π
Monotonicity	Increasing on $\left[-\frac{\pi}{2}+2k\pi, \frac{\pi}{2}+2k\pi\right]$, decreasing on $\left[\frac{\pi}{2}+2k\pi, \frac{3\pi}{2}+2kx\right], (k \in \mathbb{Z})$
Axis of Symmetry	$x = \frac{\pi}{2} + k\pi, k \in \mathbb{Z}$
Point of Symmetry	$(k\pi, 0), k \in \mathbb{Z}$
Zeros	$x = k\pi, k \in \mathbb{Z}$

Cosine curve:	$y = \cos x$
Domain	$\mathbb{R}$
Range	$[-1,1]$
Parity	Even
Periodicity	Periodic function with fundamental period of 2π
Monotonicity	Increasing on $[\pi + 2k\pi, 2\pi + 2k\pi]$, decreasing on $[2k\pi, \pi + 2kx]$, $(k \in \mathbb{Z})$
Axis of Symmetry	$x = k\pi, k \in \mathbb{Z}$
Point of Symmetry	$\left(\frac{\pi}{2} + k\pi, 0\right), k \in \mathbb{Z}$
Zeros	$x = \frac{\pi}{2} + k\pi, k \in \mathbb{Z}$

Dilation

Given the graph of $y = f(x)$,

Vertical dilation: $y = cf(x)$

Horizontal dilation: $y = f(cx)$

For $y = cf(x)$,

Vertical stretch with scale factor $|c|$ when $|c| > 1$;

Vertical shrink with scale factor $|c|$ when $|c| < 1$;

For $y = f(cx)$,

Horizontal stretch with scale factor $|\frac{1}{c}|$ when $\left|\frac{1}{c}\right| > 1$;

Horizontal shrink with scale factor $\left|\frac{1}{c}\right|$ when $\left|\frac{1}{c}\right| < 1$;

5.5 Tangent Functions

Tangent curve:	$y = \tan x$
Domain	$\{x \in \mathbb{R} \,\|x \neq \frac{\pi}{2} + k\pi, k \in \mathbb{Z}\}$
Range	$\mathbb{R}$
Parity	Odd
Periodicity	Periodic function with fundamental period of π
Monotonicity	Increasing on $\left[-\frac{\pi}{2} + k\pi, \frac{\pi}{2} + k\pi\right]$,
Axis of Symmetry	$x = \frac{\pi}{2} + k\pi, k \in \mathbb{Z}$
Point of Symmetry	$\left(\frac{k\pi}{2}, 0\right), k \in \mathbb{Z}$
Zeros	$x = k\pi, k \in \mathbb{Z}$

5.6 Inverse Trigonometric Functions

Inverse functions ($f^{-1}(x)$)

A function f has an inverse function g if and only if

(i) $f(g(x)) = x$ for all x in the domain of $g(x)$

(ii) $g(f(x)) = x$ for all x in the domain of $f(x)$

- A function only has an inverse function if no horizontal line intersects its graph at more than one point.

Properties

1. The domain and range of f Is the range and domain of f^{-1} respectively.
2. All functions must be one-to-one functions to have an inverse function.
3. The graph of f^{-1} is the reflection of the graph of f about $y = x$.

4. $\boldsymbol{f\left(f^{-1}(x)\right)=x \Rightarrow x \in D^{-1}, f^{-1}(f(x))=x \Rightarrow x \in D}$

Inverse trigonometric function	$y = \arcsin x$	$y = \arccos x$	$y = \arctan x$
Domain	$[-1,1]$	$[-1,1]$	$\mathbb{R}$
Range	$\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$	$[0, \pi]$	$\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$
Parity	Odd	Neither	Odd
Monotonicity	Increasing	Decreasing	Increasing

Identities

1) $\sin(\arcsin x) = x, x \in [-1,1]$; $\cos(\arccos x) = x, x \in [-1,1]$; $\tan(\arctan x) = x, x \in \mathbb{R}$;
2) $\arcsin(\sin x) = x, x \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$; $\arccos(\cos x) = x, x \in [0, \pi]$; $\arctan(\tan x) = x, x \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$.

5.7 Trigonometric Equations

Equation	General solution
$\sin x = a, \lvert a\rvert < 1$	$x = k\pi + (-1)^k \cdot \arcsin a, k \in \mathbb{Z}$
$\cos x = a, \lvert a\rvert < 1$	$x = 2k\pi \pm \arccos a, k \in \mathbb{Z}$
$\tan x = a$	$x = k\pi + \arctan a, k \in \mathbb{Z}$

Type 1:

$$a \sin^2 x + b \sin x + c = 0 \ (a \neq 0)$$

Type 2:

$$a \sin x + b \cos x + c = 0 \ (a^2 + b^2 \neq 0, c \neq 0)$$

Type 3:

$$a \sin x + b \cos x = 0 \Leftrightarrow a \tan x + b = 0$$

$$a \sin^2 x + b \sin x \cos x + c \cos^2 x = 0 \Leftrightarrow a \tan^2 x + b \tan x + c = 0$$

$$\Leftrightarrow A \sin 2x + B \cos 2x = C$$

Type 4:

$\sin f(x) = \sin \phi(x)$	$f(x) = 2k\pi + \phi(x)\ or\ 2k\pi + \pi - \phi(x), k \in \mathbb{Z}$
$\cos f(x) = \cos \phi(x)$	$f(x) = 2k\pi \pm \phi(x), k \in \mathbb{Z}$
$\tan f(x) = \tan \phi(x)$	$f(x) = k\pi + \phi(x), k \in \mathbb{Z}$

Type 5:

-substitution method can be used for equations with $\sin x \pm \cos x$ and $\sin x \cos x$

Let $\sin x \pm \cos x = t$ ($|t| \leq \sqrt{2}$). $\sin x \cos x = \pm\frac{t^2-1}{2}$.